



**Politecnico
di Torino**

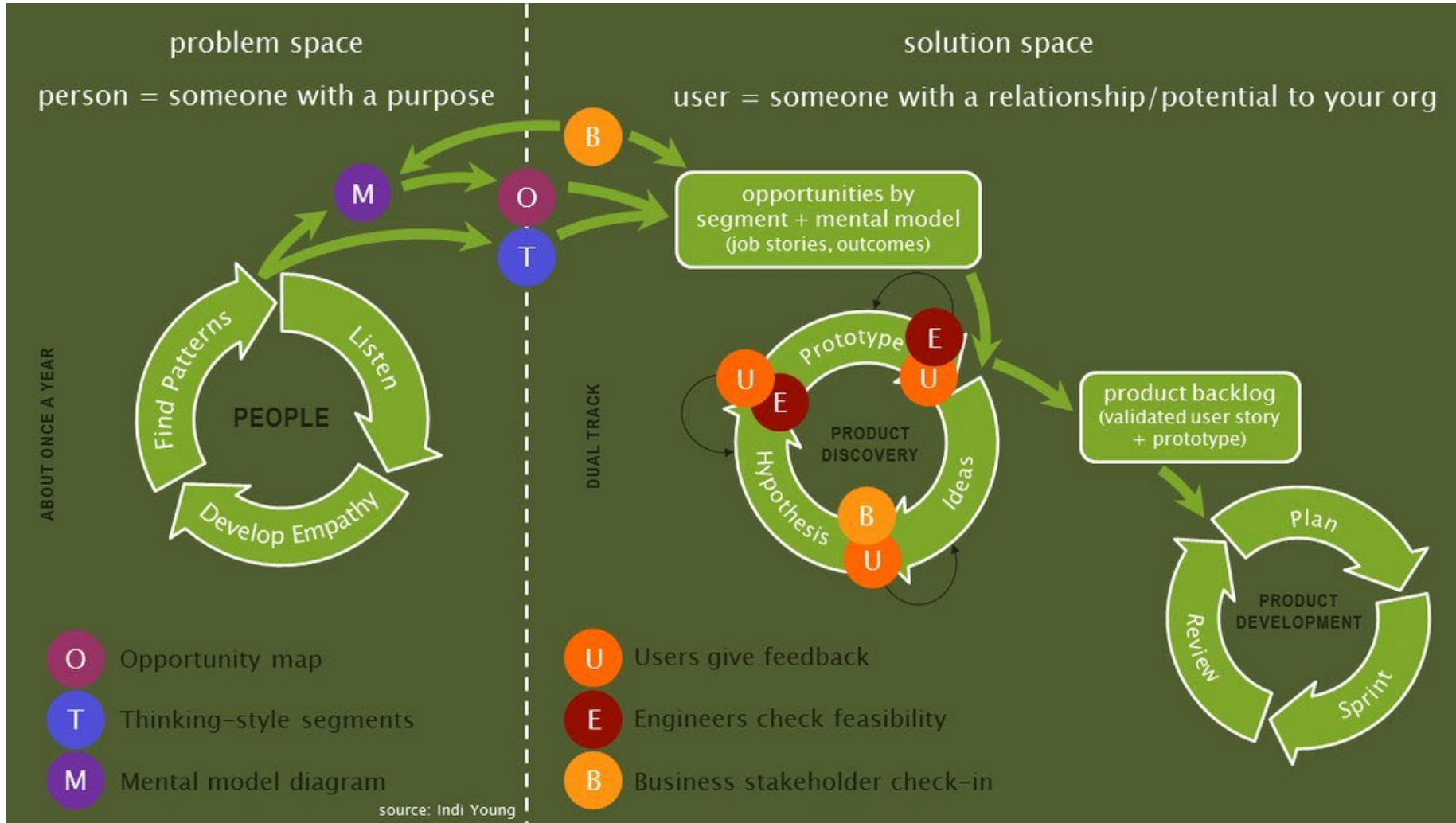
Applied Data Science Project

User Personas

and other design thinking tools

Antonella Frisiello

Learn from people perspectives to set the solution.



Learn before designing. Prototype before developing.

DATA COLLECTION



Assume a
Beginner's Mindset



Photo- & Video-Based
Observations



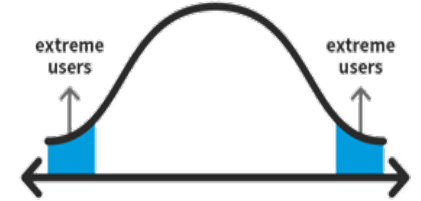
Personal Photo
& Video Journals



Conduct Interviews
with Empathy



Bodystorm



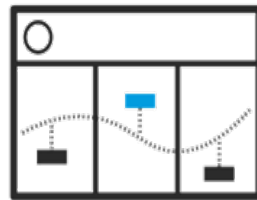
Engage with
Extreme Users

+

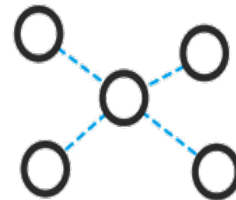
DATA DISPLAY



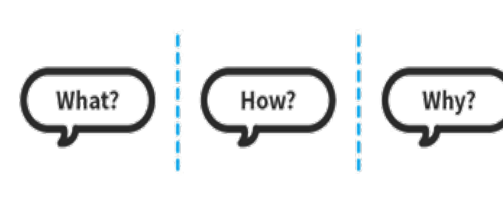
Personas



Journey Mapping



Embrace Analogies



The What-How-Why Method



Capture & Share
Inspirational Stories



User personas

[UX Visualization Examples & Tips](#)



User Personas

Personas are **data-driven portraits** of **archetypical stakeholders** that help designers and developers focus on the needs and goals of target users throughout the product development process.

that represent the key attributes, actions, behaviours, needs, aspirations, difficulties and frustrations of the segments to which the project is aimed (*).

They are not the portrait of a specific subject, but the **synthesis of the primary data** of a relevant group of subjects, different for socio-demographic variables, roles, OBJECTIVES, contexts of use...

They are a **data-driven synthetic representation** of complex knowledge, useful for internal communication...

They are **dynamic documents**, to be updated iteratively, during the project.



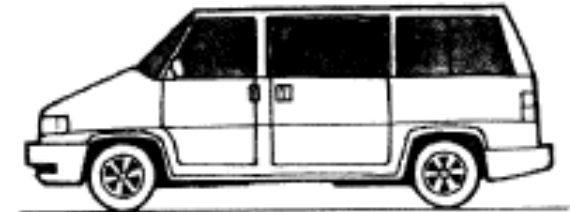
Alessandro's goals

- Go fast
- Have fun



Marge's goals

- Be safe
- Be comfortable



Dale's goals

- Haul big loads
- Be reliable



Figure 5-2: A simplified example of how personas are useful. By designing different cars for different people with different specific goals, we are able to create designs that other people with similar needs to our target drivers also find satisfying. The same holds true for the design of digital products and software.

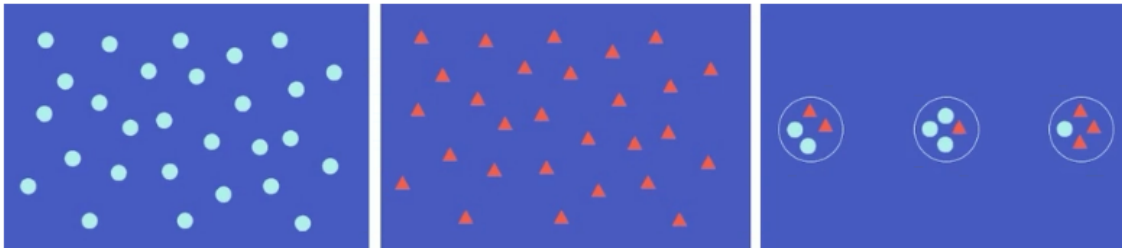
Alan Cooper. [About Face: The Essentials of Interaction Design 2.0](#).

Example

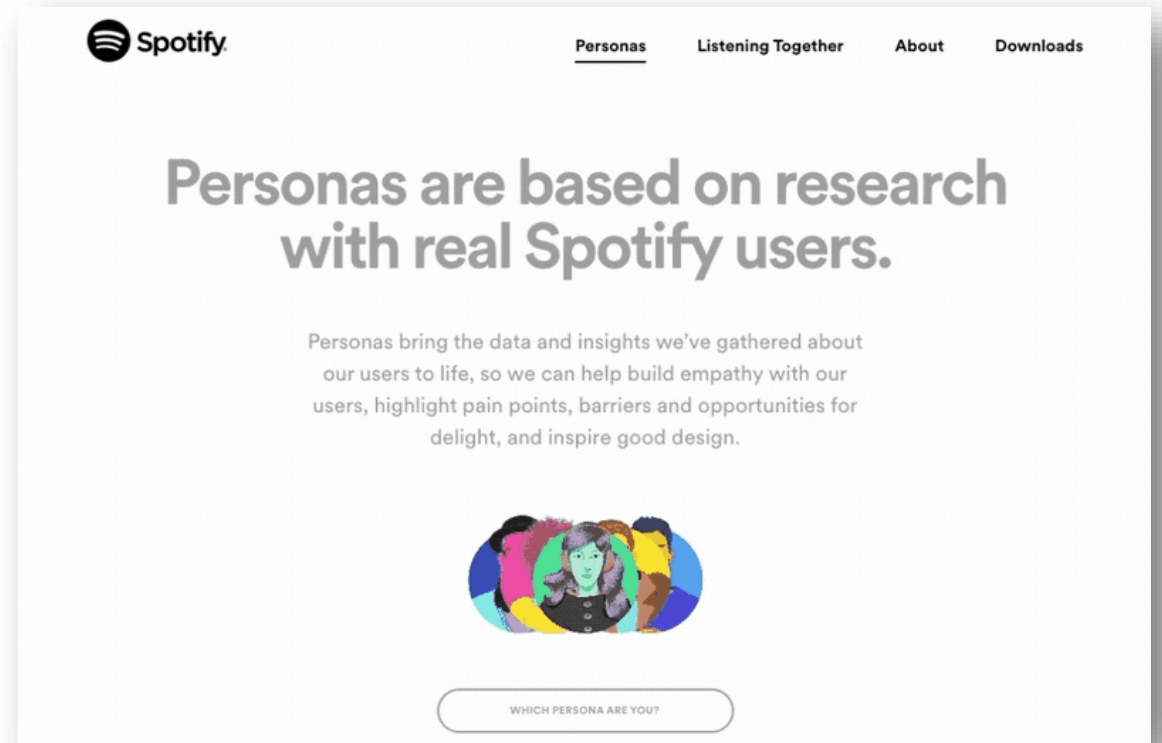
Phase 1 (2017)

Phase 2 (2018)

The initial challenge was to overcome the idea of a unique audience.
Then put a face to listeners to facilitate discussion and empathy at the company level.



Personas combine similar user needs, habits, and attitudes and communicate the nuanced commonalities and differences between our users.



The visual representation used 2D illustrations with brand colours to keep them abstract but recognisable.

Personas represent the «target group» for which it is designed, making it concrete and real, articulated.

Personas are empathetic design tools to **delve deeper into real needs** and keep them in mind throughout the process

Personas are inclusive: despite being data-driven, they facilitate those who design and develop in taking into consideration the needs of minority groups, improving the general solution.

PERSONA PROFILE



INTERESTS:

What does she spend her time on?
What enlivens her?
What does she read, watch, listen to?

POWERS:

What Resources does she control?
What can she do, muster?
Who does she have influence over?

Name:
Fictionalize it
Profession:
Be very specific

Age:
Choose a number,
Not a Range

Personal
Bg:
Mini Life story -
Hometown, Family,
Schooling, Work, etc

NEEDS:

What does she need,
in her life, in her work?
What's essential for her?
What does she require?
What's missing?

BEHAVIORS:

What are her
ROUTINES
BUYING
Free Time
Rituals

VALUES:

What carries meaning for her?
What's her moral compass?
What makes her indignant,
satisfied, frustrated?
What does she want for the world?

ASPIRATIONS:

What kind of person
do they want to be?
Brands?
Spending patterns?
Life dreams? Heroes?
Role models?

The User Personas

PRO: PERSONAS



- enable designers and developers to **empathise** with these imaginary users and **understand better** their goals and needs, taking on the perspective of underrepresented or easily overlooked users.
For example, Personas explicitly designed to represent **human diversity**, known from gender difference research, have been successfully used to detect gender-inclusiveness issues in software.
- **aid the communication within the team** (i.e., the term 'user' can mean different things to different people within the group), making implicit assumptions of users explicit.
- support the communication with the stakeholders, adopting **a common language about users**
- closely **approximate the mental model of various end users**, which results in a focus on the user priorities and a meaningful feature set.
- Help to **avoid self-referential projects** based on the user's (and not developers') goals, motivations, and skills.

CONS: PERSONAS



- are prone to activate and **reinforce stereotypes**.
For this reason, it is necessary to ensure that the diversity of people is accounted for. The way to do it is monitor the data quality.

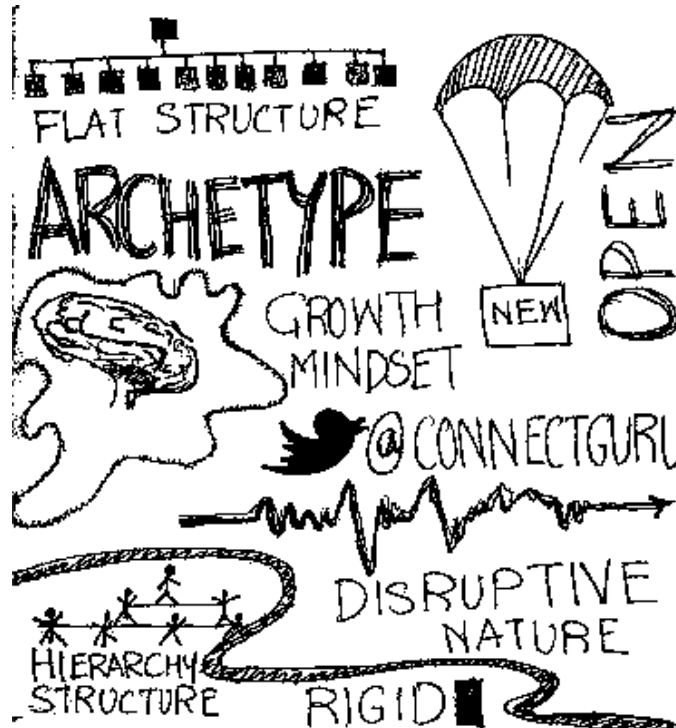
Stereotype vs archetype

Archetypes are universal symbols or **patterns** that are found in stories and data. They represent fundamental human experiences and motivations, used to plan social media campaign and marketing strategies.

The **Hero** is a character who embarks on a journey, faces challenges, and...

The **sage**: Yoda in "Star Wars."

The **Trickster**: Loki in Norse mythology.



Stereotypes are over simplified, generalised ideas about a group of people. They can be positive or negative, but they are often **based on inaccurate or incomplete information**.

- stereotypes can lead us to make not accurate assumptions (e.g. all gamers are young, male and antisocial)

Teenagers are rebellious and irresponsible.

Women are bad drivers

All Italians are passionate and loud.



Again the categories.

The world doesn't need categories. It's human beings who need them. **We build categories to navigate this complex and contradictory world**, to somehow understand it and come to an agreement on it.

We need categories.

Anyone who tried to perceive everything on this earth without filters and without categories would be overwhelmed by a flood of stimuli and would end up drowning. **We need categories**. Organising and categorising our environment helps us **to identify patterns, make quick decisions, and react to situations of danger**, for example. In these moments, we draw on images and situations we have stored long before.

So, **looking at the world through categories is a necessity**. But when do the categories we build to understand the world **become cages**? When does our freedom become a lack of freedom for others? It is faith in absolutes that transforms categories into cages - the arrogant belief that a single, narrow, and limited perspective on this world is complete, exhaustive, and universal. The hubris of believing we can fully understand another human being in their complexity once and for all. Or even that we've definitively understood an **entirely artificial category of human beings**.

More than 70 million human beings become THE REFUGEE.

1.9 million people become THE MUSLIM.

Half the world's population becomes THE WOMAN.



Which stereotypes can affect your project?

What categories are **represented** in your dataset?

How was the data collected?

Are any key typologies **underrepresented**?

Could the collection method exclude certain information?

Project goals	Possible stereotypes	Background	Possible Personas
Predictive model for Humanitarian Aid	Immigrants primarily hold low-wage jobs and hurt the economy”	Germany calls for more immigrants to fix its shrinking economy, Financial time , Reuters (2023)	Employers, Employee, Candidates
Urban Air Quality Assessment and High-Resolution Temperature Mapping	Non-experts rely on basic sensory cues to perceive air quality, such as visibility, smell, or feel; neighbourhood halo effect;	Air pollution perception bias: Mismatch between air pollution exposure and perception of air quality in real-time contexts (2019)	Citizens, CEO of industries, Physicians, Public officers
Heat Forecast	People don’t see heatwaves as a pressing issue.	5 things we’ve learnt about people’s experiences of heatwaves (2018)	Frail citizens (elderly, pregnant, children, chronic patients), Red cross and responders Physicians, Public officers
Pre-training language model on Electronic Health Records Incorporating Patient Preference Studies into clinical research and decision models	Black patients are more likely than White patients to be described in negative terms	Electronic Health Record Notes May Be Perpetuating Biases (2022)	Patients (various), Physicians, Medical institutions, Insurance companies
Unsupervised topic detection in app reviews	Users’ feedback may be dismissed as biased or emotional rather than informative.	Topic modeling helps extract insights from user-generated content without manual labeling (Tushev, M., Ebrahimi, F., & Mahmoud, A. (2022).	Product managers, Data scientists, App users, Marketing teams
AI Personas	Personas might reinforce gender, racial, or occupational stereotypes if generated without bias control.	Synthetic personas in design can amplify demographic bias (Shin et al., 2024).	Marketing teams, AI developers, End users

Dataset Examination

1) Explore your dataset to identify potential stereotypes:

Search for missing data or underrepresented variables and stereotypes in:

- **sampling**
- **data collection**
- **labelling**

2) Search for this information:

- **Who** created the content?
- **What** is the content?
- **When** was the content created?
- **How** was the content created?
- **Why** was the content created?
- **What** do they count?
- **What do these data not tell us?**

3) Check for representation across different stakeholders and demographic groups.

- Are there **features that could unfairly disadvantage certain groups** or individuals or cause risks or unwanted effects?
- Have you added features based on assumptions that might not hold for all data points?
(E.g., including a zip code as a feature in a model could introduce socio-economic or racial bias if specific zip codes correlate strongly with demographics).

4) Describe the possible impact on the project and outcomes

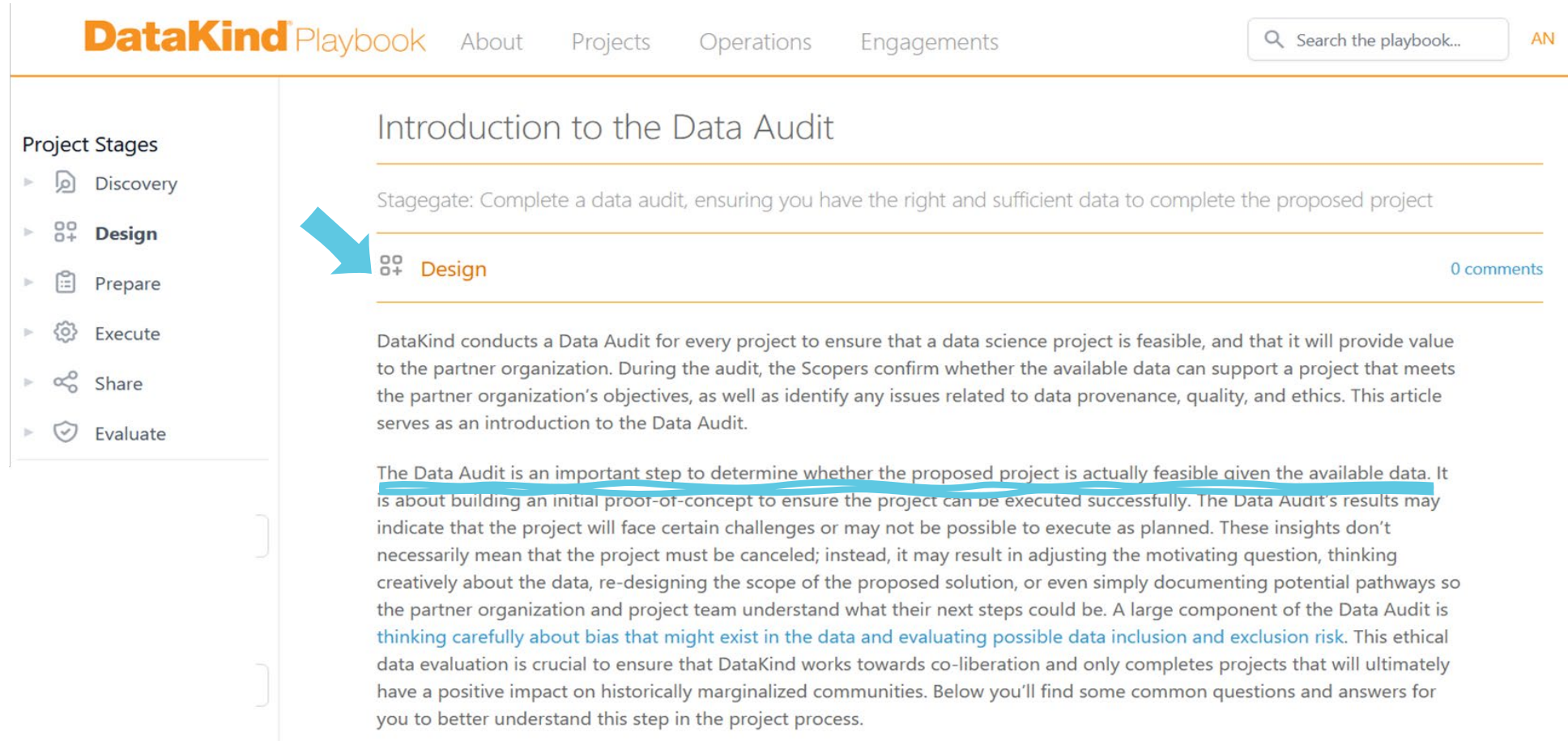
5) Identify actions to mitigate the identified biases.

- **Are your metrics fair?** How does the model performance differ across various groups?
(E.g. if accuracy is your primary metric, but accuracy varies considerably between groups, then accuracy might not be the only metric to consider)

Data audit toolkit

A Six-Step Playbook for Doing Data Science & AI for Good

- Identifying problems
- Understand the data
- Build proof-of-concept models
- Implement models at scale
- Evaluate the project's ethical implications



The screenshot shows the DataKind Playbook website. The header includes the 'DataKind Playbook' logo and navigation links for 'About', 'Projects', 'Operations', and 'Engagements'. A search bar is located on the right. On the left, a sidebar lists 'Project Stages' with icons and labels for 'Discovery', 'Design' (highlighted with a blue arrow), 'Prepare', 'Execute', 'Share', and 'Evaluate'. The main content area is titled 'Introduction to the Data Audit' and includes a 'Stagegate' description: 'Complete a data audit, ensuring you have the right and sufficient data to complete the proposed project'. Below this, the 'Design' section is introduced, stating that DataKind conducts a Data Audit for every project to ensure feasibility and value. A paragraph is highlighted in blue, discussing the importance of the Data Audit in building a proof-of-concept and evaluating data inclusion and exclusion risks. The page also shows '0 comments'.

DataKind Playbook About Projects Operations Engagements

Search the playbook... AN

Project Stages

- ▶ Discovery
- ▶ **Design**
- ▶ Prepare
- ▶ Execute
- ▶ Share
- ▶ Evaluate

Introduction to the Data Audit

Stagegate: Complete a data audit, ensuring you have the right and sufficient data to complete the proposed project

Design 0 comments

DataKind conducts a Data Audit for every project to ensure that a data science project is feasible, and that it will provide value to the partner organization. During the audit, the Scopers confirm whether the available data can support a project that meets the partner organization's objectives, as well as identify any issues related to data provenance, quality, and ethics. This article serves as an introduction to the Data Audit.

The Data Audit is an important step to determine whether the proposed project is actually feasible given the available data. It is about building an initial proof-of-concept to ensure the project can be executed successfully. The Data Audit's results may indicate that the project will face certain challenges or may not be possible to execute as planned. These insights don't necessarily mean that the project must be canceled; instead, it may result in adjusting the motivating question, thinking creatively about the data, re-designing the scope of the proposed solution, or even simply documenting potential pathways so the partner organization and project team understand what their next steps could be. A large component of the Data Audit is thinking carefully about bias that might exist in the data and evaluating possible data inclusion and exclusion risk. This ethical data evaluation is crucial to ensure that DataKind works towards co-liberation and only completes projects that will ultimately have a positive impact on historically marginalized communities. Below you'll find some common questions and answers for you to better understand this step in the project process.

Creating personas

There is no single way of creating and using personas, neither in literature nor in practice.

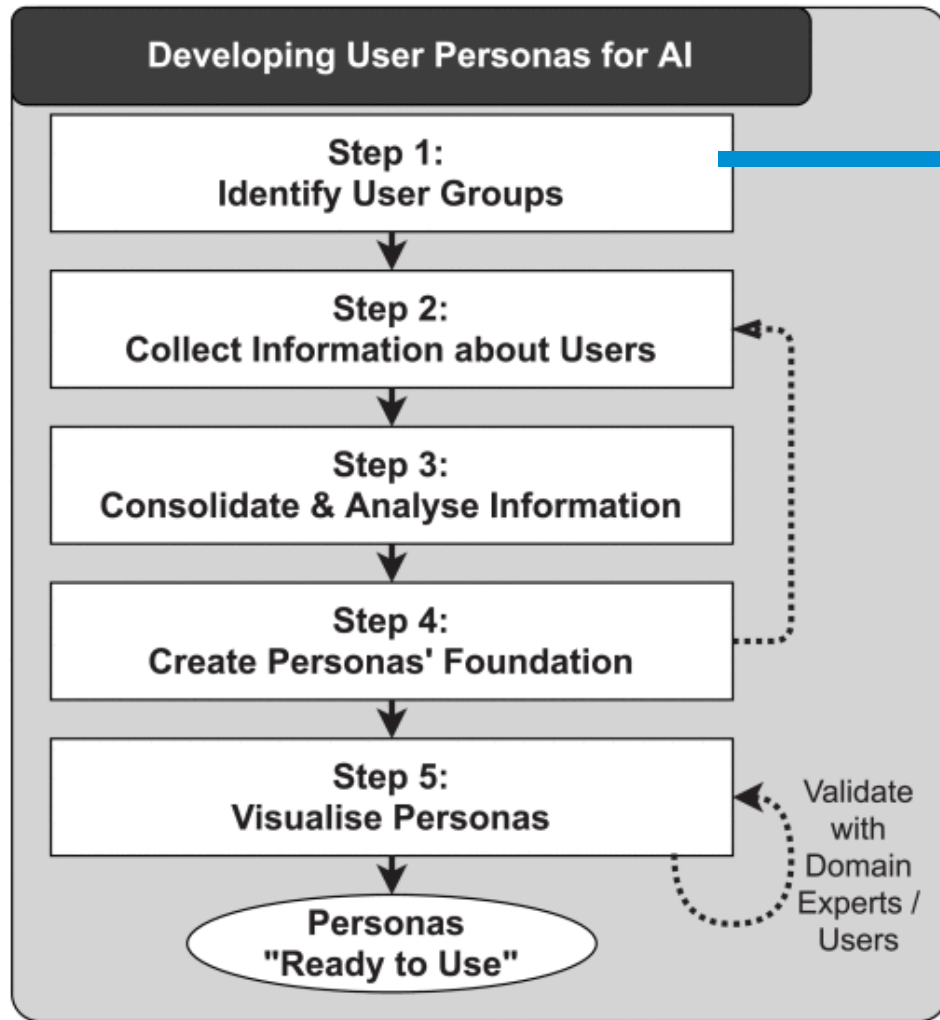
Usually, with the exception of pastiche personas, **personas are based on data and information collected about real people** by using **qualitative methods** such as ethnographic interviews, open-ended survey questions, or contextual inquiries and field studies.

In the last 15 years, a collection of **large amounts of quantitative data** (for example, from web analytics, social media, online customer data, and online surveys) together with machine learning techniques led to so-called **digital data-driven personas** (mainly used in marketing and customer research).

So-called **hybrid personas** are created by utilising quantitative data from online analytics and qualitative insights.

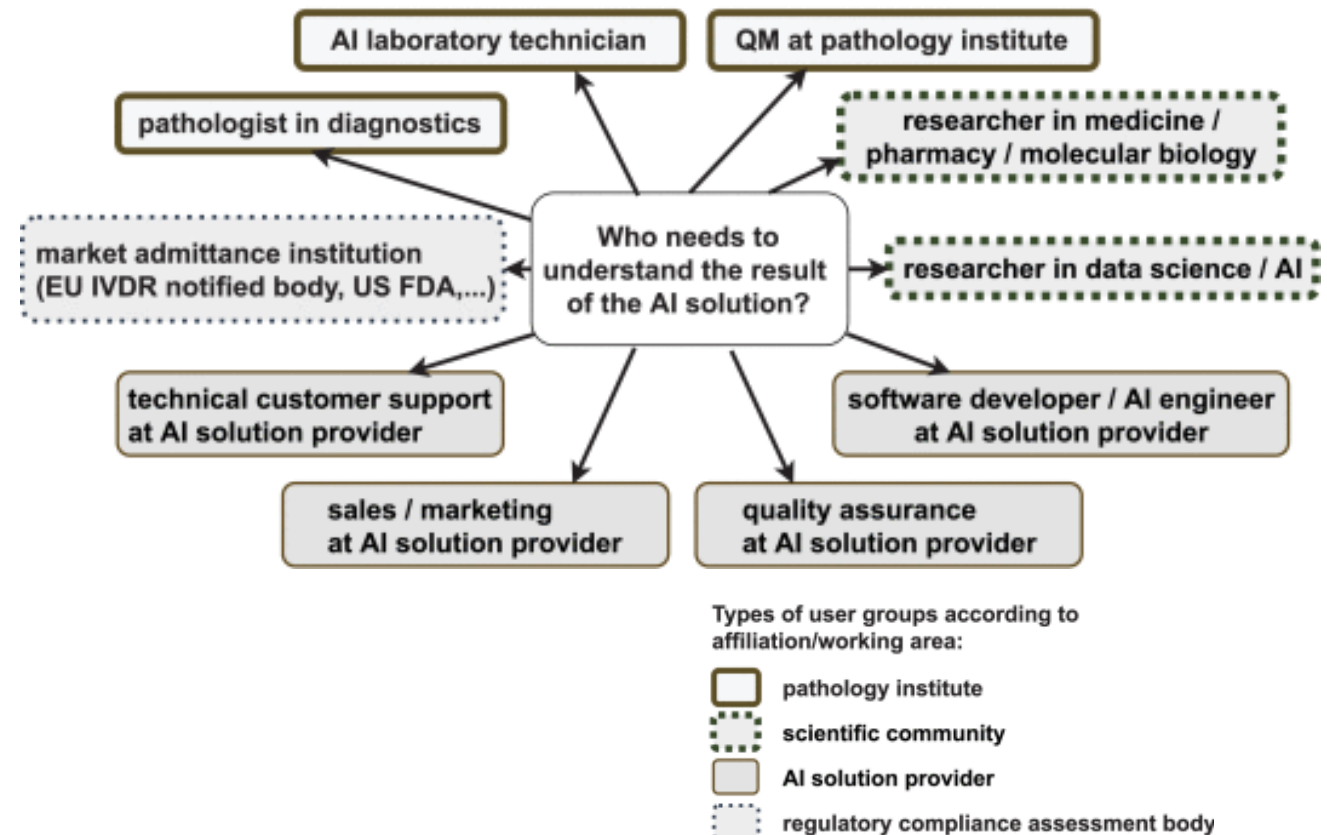


Design Personas for AI

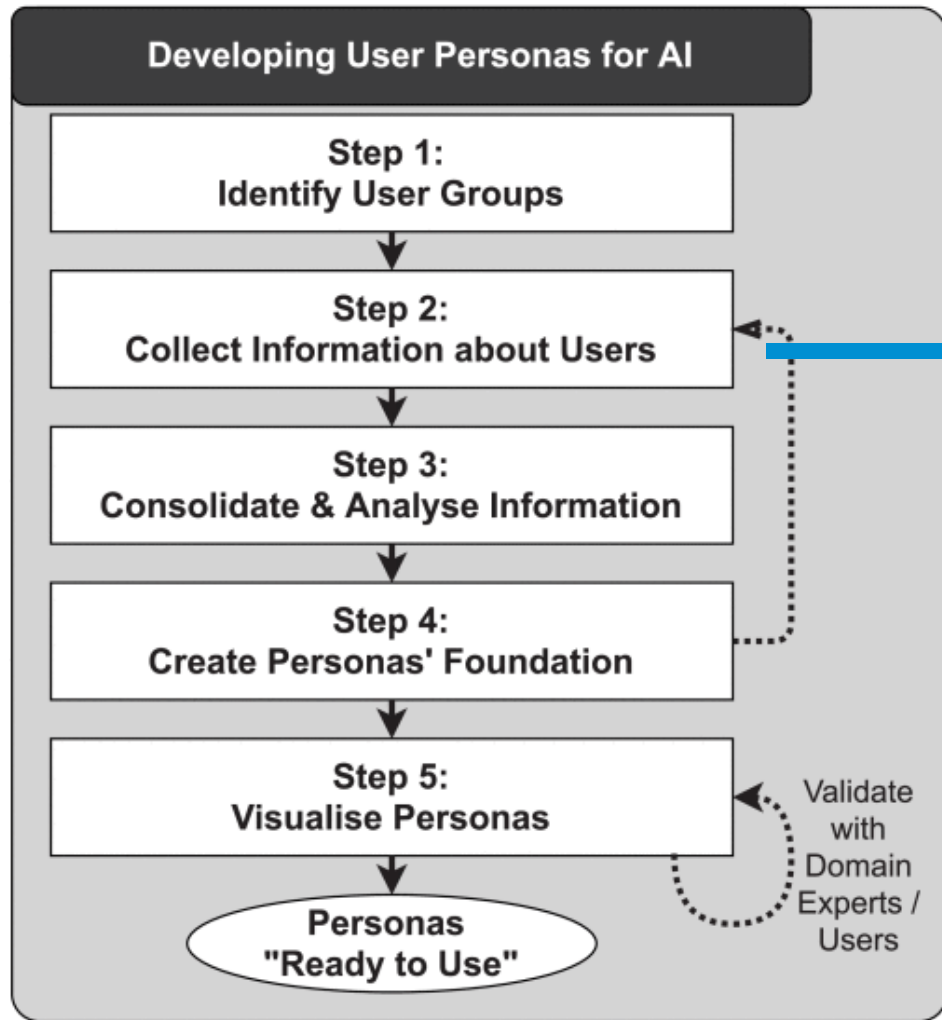


STAKEHOLDERS MAP

User Groups for AI Application in Computational Pathology



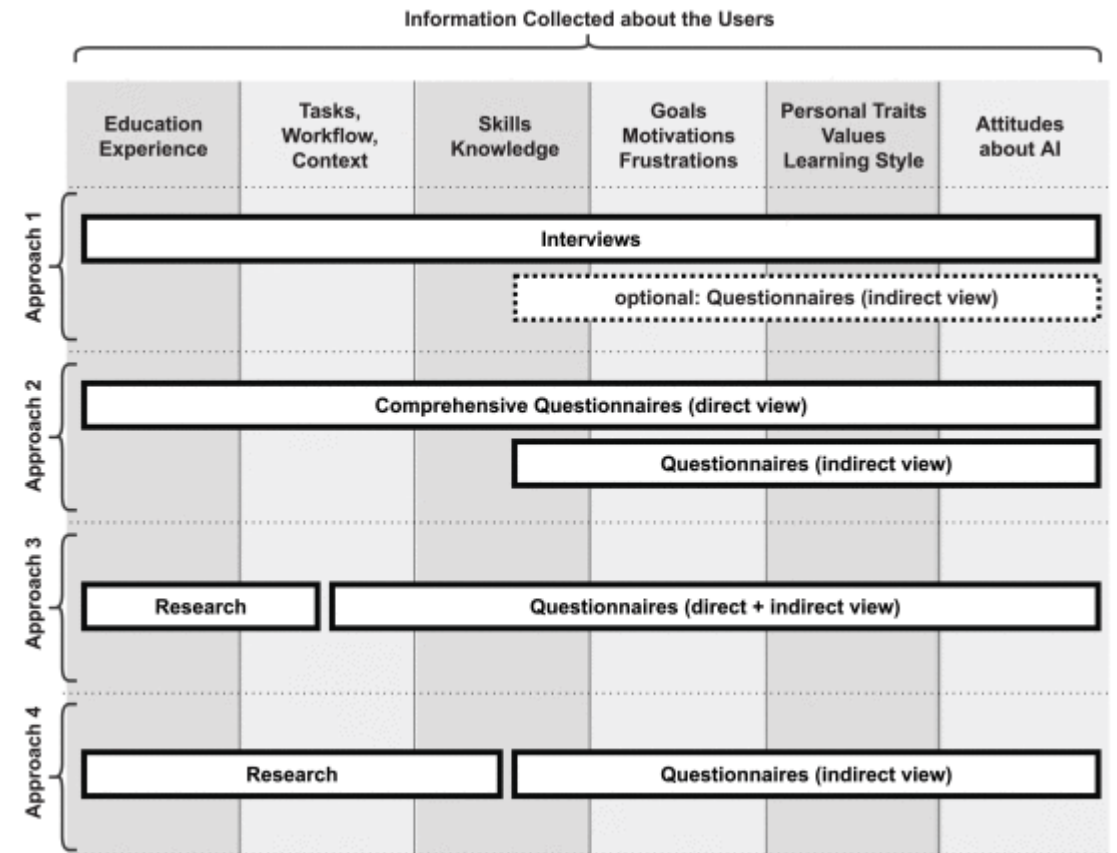
Design Personas for AI



What categories are **represented** in your dataset?
Are any key typologies **underrepresented**?
How was the data collected?
Could the collection method exclude certain information?

What to search → know?

- **Demographics:** age, gender, geographical location of the users.
Sources: social media analytics, website analytics, previous research, competitive research.
- **Motivation:** why are the users using our product?
Sources: previous research, competitive research, first-hand experience with users, scientific literature.
- **Goals:** short-term and long-term goals (personal/career), tasks and activities to reach them. Sources: the same used for motivation, psychographics, and skills.
- **Skills:** education, relevant skills and professional experience. Any technical skills as well as soft skills, *services/systems already in use*.
Sources: previous studies, surveys, first-hand experience with our users, external research.
- **Psychographics:** personality types, values, opinions, beliefs, attitudes, activities, interests, lifestyles, etc.
Sources: social media analytics, industry publications, blogs, academic research.

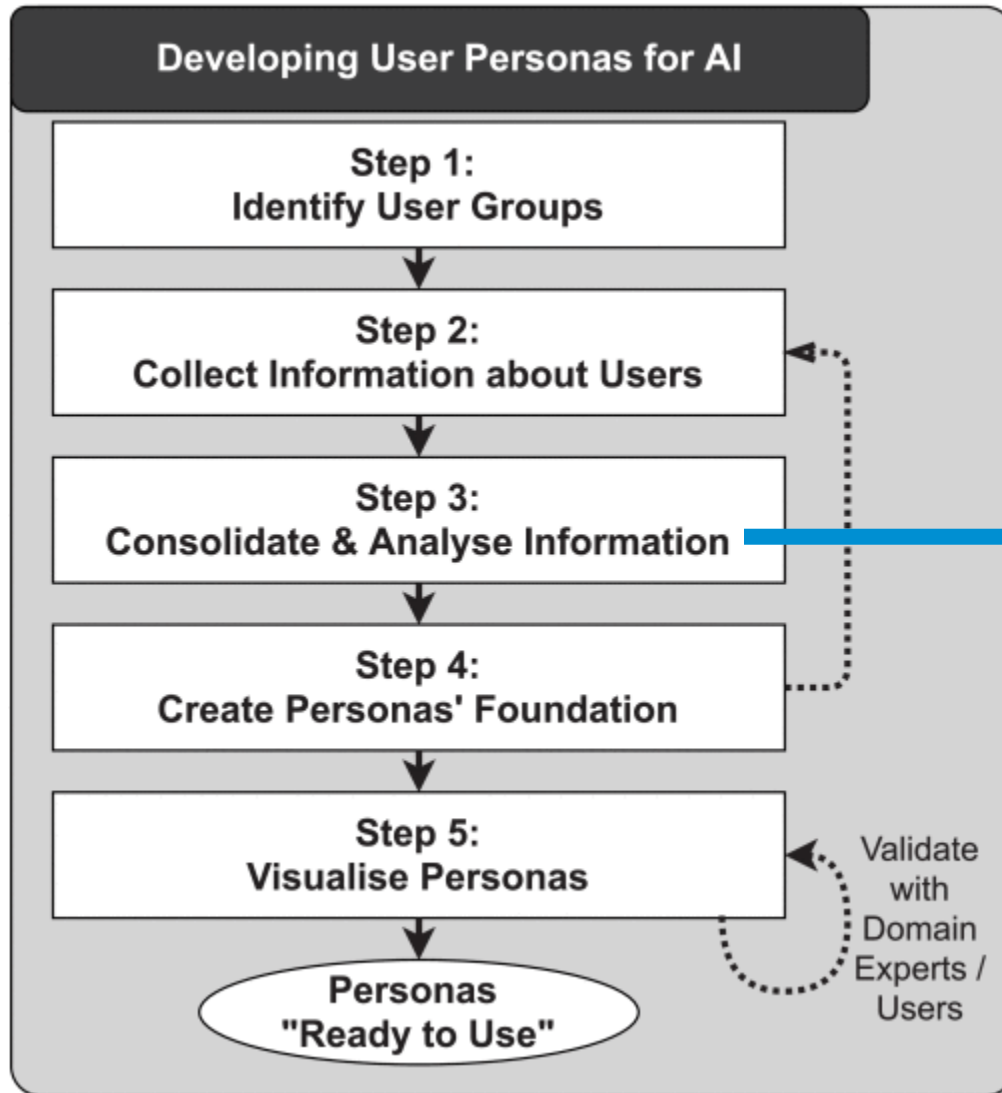


NN/g

Statistical Personas



Design Personas for AI

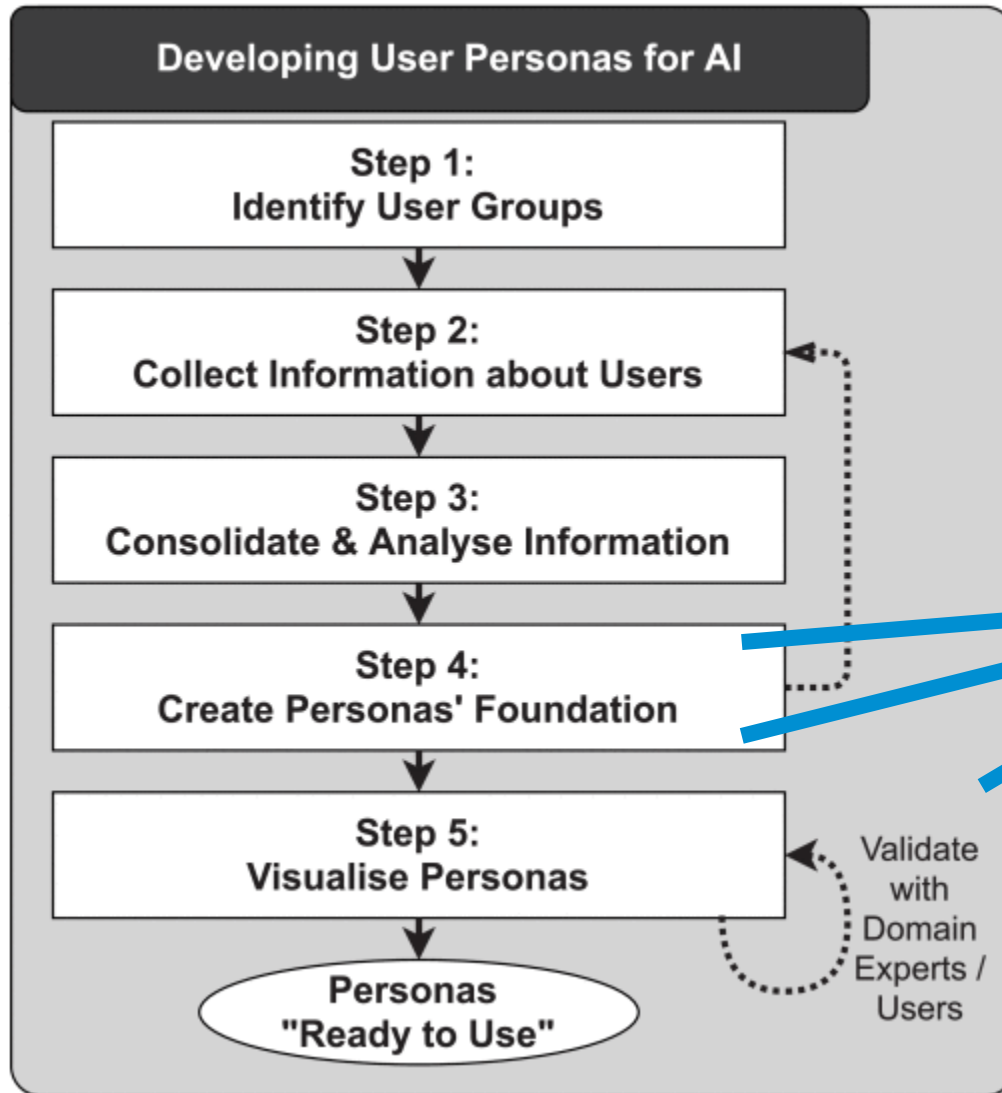


Pruitt, J. & Adlin, T. The Persona Lifecycle: Keeping People in Mind Throughout Product Design . Morgan Kaufmann, 2006.

Kreitzberg, C.B., Little, A. Usability in Practice - [The Power of Personas](#) Microsoft Ignite, Volume 24 Number 04



Design Personas for AI



CREATE THE USER PERSONAS

Per every personas include:

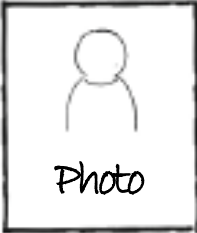
- A **photo**/portrait
- **Short bio** /demographic details
- **Mindset**: a brief description of the point of view (specific and characteristic of the person in relation to one or more aspects of the project theme)
 - Role, Tasks, values
 - Work (tasks, workflows, context)
 - Education/knowledge/skills
- **Goals**, values
- Relevant **pain points and gains, motivations and barriers**
- Attitudes towards **technology/AI**

Personas I one template

Fictional but realistic photo based on data
(thispersondoesnotexist.com)



Short-bio:
relationship with the
main life/work
context in which it
acts (and object of the
project)

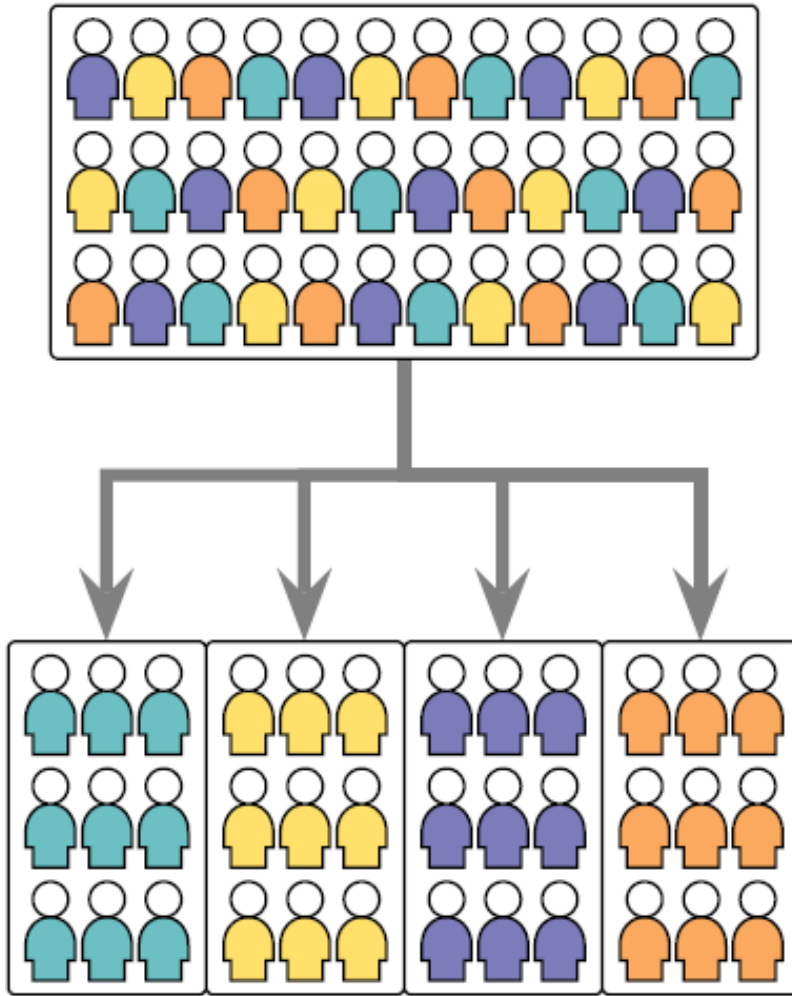
PERSONA		DATE	
 Photo	Name Role Socio-demographics	Fictional but realistic name Role	Quotation
	Difficulty: barriers that it faces or poses	Data literacy / digital skills	Literacy/ confidence level with data e and tech relevant for the project (e.g. basic, intermediate, advanced/pro)
Short bio	Needs	Goals	Barriers
	Needs: necessity, difficulty s/he faces	Objectives: what s/he wants to achieve and why (motivations) Aspirations: desired solutions	Difficulty s/he faces or poses

Date of creation or last update
(yes: Personas must be updated)

A representative phrase of the character.

Literacy/ confidence level with data e and tech relevant for the project (e.g. basic, intermediate, advanced/pro)

Guarantee representation and diversity



- **Diversity:** inclusion of a wide range of perspectives, experiences, and identities (including race, gender, age, socioeconomic status, disability, culture, and more)
- **Representation:** visibility and inclusion of all groups, especially those historically marginalised. if you don't count them, they do not exist.
- **Intersectionality:** framework used to understand how various variables (ethnicity, gender, age, education, ability, and others) **intersect and overlap** in more complex scenarios.

Personas types

Usually, when creating one Persona, **not the whole person is described**, but the focus is put only on **relevant aspects** (goals, needs, attitudes, skills...) and the **specific context** associated with these aspects. Personas usually encompass aspects such as **context and environment, tasks and workflows, skills and knowledge, personal traits, goals, values, motivations but also frustrations**.

- **Goal-oriented personas**, distinguished from one another based on their different goals (Cooper, 1999).
- **Role-based personas**, defined by their roles, (Pruitt et al, 2000)
- **Scenario-oriented engaging personas**, based on their individual characteristics, and their goals are based on these needs and appear only in the context of a specific scenario (Nielsen).
- **Pastiche personas**, fictional portraits established on user data but entirely grounded on fictional characters from literature or film (Blythe).

To adapt the personas method to the context of HAI, additional aspects describing the user's attitude regarding AI solutions as specifically relevant for personas for AI:

- **Trust** (How much trust does the user have in the decisions/output of the AI system?)
- **Acceptance** (Does the user accept (and follow) the decision of the AI system?)
- **Assent** (Is the user willing to accept/use the support by the AI system?)

Example

Instructions

- Length of instructions for both forms = 54 pages
- These hold instructions for every type of green card applicant, intermixed so an applicant has to seek what's relevant to them

Forms

- In total the two necessary forms = 364 questions
- Many of which are situation-specific or occur twice between forms
- Minimal clarification & guidance

Document Checklist

- Minimally informative with many conditional statements (if applicable), making many of it potentially erroneous
- Too broad - missing certain important details

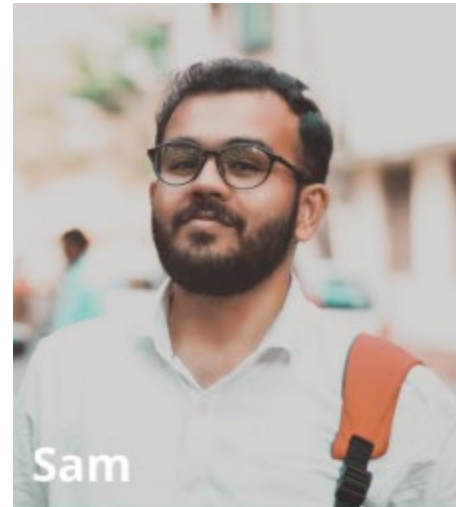
Case Status

- Case status tracker shows no information about the timing of the future steps

Tools & Resources

- Available only if a user signs in (i.e., yet the resources are not specific to the applicant's situation, requiring sifting

Desk analysis + Interviews



Sam

AGE	32
OCCUPATION	Data Scientist
STATUS	Married
LOCATION	Chicago, IL

"You really have to pay close attention and hunt for what details are actually relevant to you. What category do I fall in? You have to read it ten times over to figure it out and it can become overwhelming."

BIO

Sam was born in India but he's been a resident of the US for over 10 years. He originally immigrated on a student visa, then received a work-sponsored H1B visa; however, Sam's visa expires soon and he has no plans to go back to his hometown.

A few months ago, Sam married his long-time partner. The notice for his visa's expiration has catalyzed Sam's interest in applying for a family-based green card.

Sam has a couple of friends who have gone through this process before without a lawyer, so he feels confident he can do the same.

Sam and his partner are busy with work and other responsibilities, so they don't have the extra energy required to carefully discern this complex application process. In addition, since Sam's visa expires soon, it will place him out-of-status, which means he needs this process to progress smoothly.

The combination of the lack of time to get the application right and the pressure of needing it done as soon as possible has been frustrating for Sam, so he is seeking an affordable tool to help him out.

GOALS

- To minimize stress during application process
- To be able to continue living in the US with his spouse
- To receive approval as quickly as possible

NEEDS

- To feel in control of the outcome
- To understand the details of his application process
- To stay updated on the status of his case

FRUSTRATIONS

- Too much time spent researching legal jargon
- USCIS's use of mail for updates & communication
- Having application delayed because of insufficient supplemental documents

Example

**Marie — The artist**

Looking to sell her art easily online and gain exposure as an up and coming artist.

Age: 26

Marital status: Single

Occupation: Artist/Art Buyer

Location: London, UK

Income: £32,000

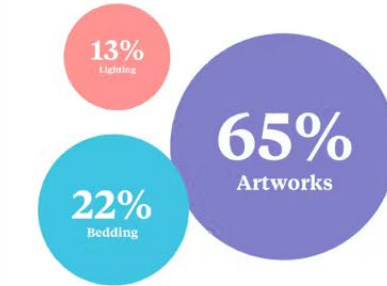
Bio

Julia is an artist in London. She graduated 4 years ago from art school and has been making art since joining a local studio 12 months ago.

Although she loves the people she works with, she has been giving serious thought to her future. While the studi that she works at is growing, she's worried that she wont developed a name for herself if she continues working with others. She's considering selling her art online and wants a reputable place to sell her pieces while gaining good exposure.

Needs

- To find the right website which offers the facilities she wants.
- To find a way to fund her artworks without incurring lots of debt.

**Pain points**

- Concerned that she'll need to manage mailing artworks and won't be able to afford the website fees
- Worried that she's one of many artists and wont get enough exposure
- Doesn'twant the service to take too high a percentage of her sales

Ideal experience

- Mail her pieces internationally and ensure they arrive safely
- Manage her inventory easily from her iPhone
- Make great money to sustain her passion making art

Research

2 cards, 1 document

Quotes

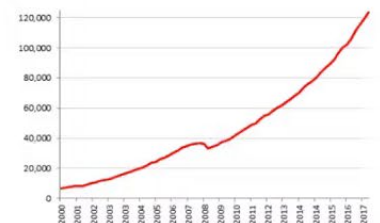
"It's important that I can connect with the buyers"

"I really just need an easy way to promote my art and keep prospective buyers up to date"

"The hardest part is managing the actual transaction."

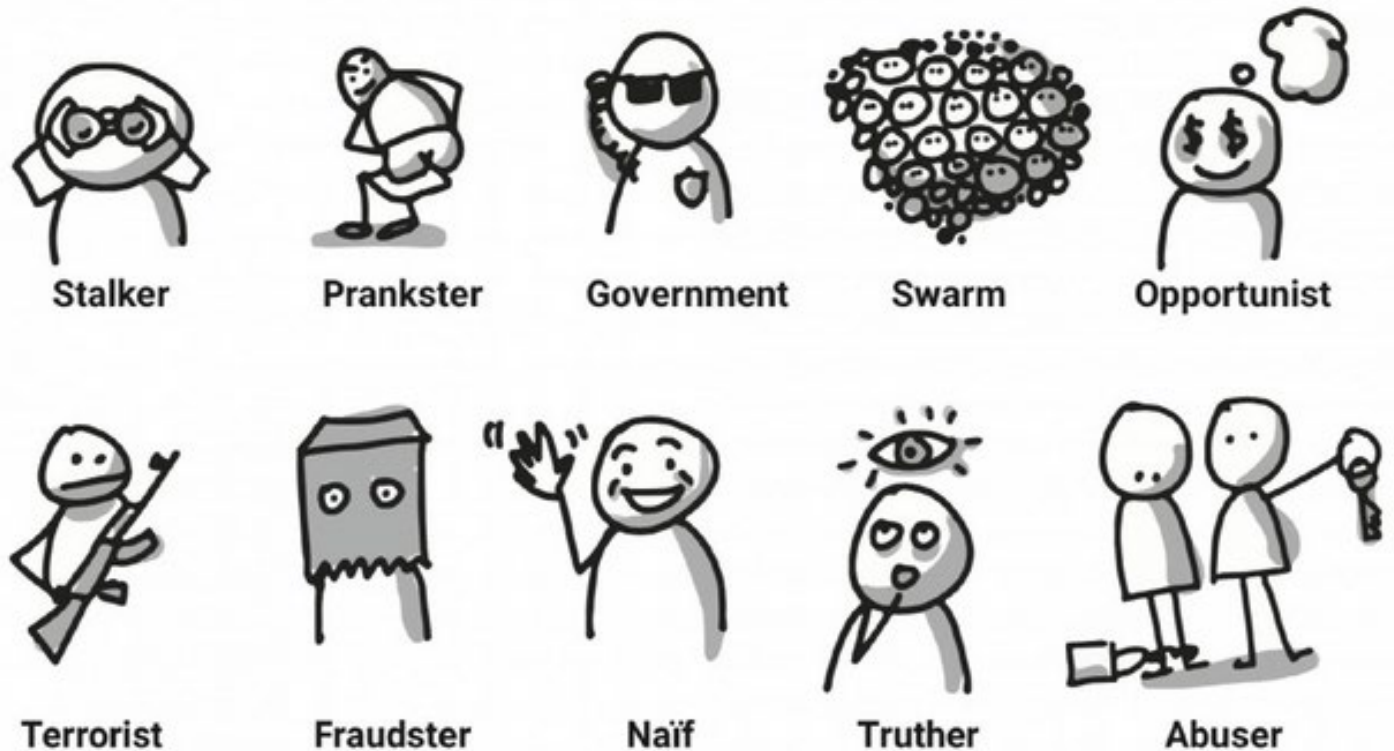
**Survey results**

3 words

E-Commerce Sales

Dark personas

When designing, we are entirely focused on actions and behaviours we want to foster and support.
But there will always be someone ready to use our service in **unexpected and unwanted ways**.



Tweet di @mollyclare

Synthetic personas

LLMs can be used to create Personas based on real or textual data.

- Advantages: speed, upgradeability, consistency.
- Risks: bias, stereotypes, lack of authenticity and empathic depth.

Empirical validity:

the insights generated must always be validated with real users.
It must be used in a transparent, ethical and complementary way.

USE CASES

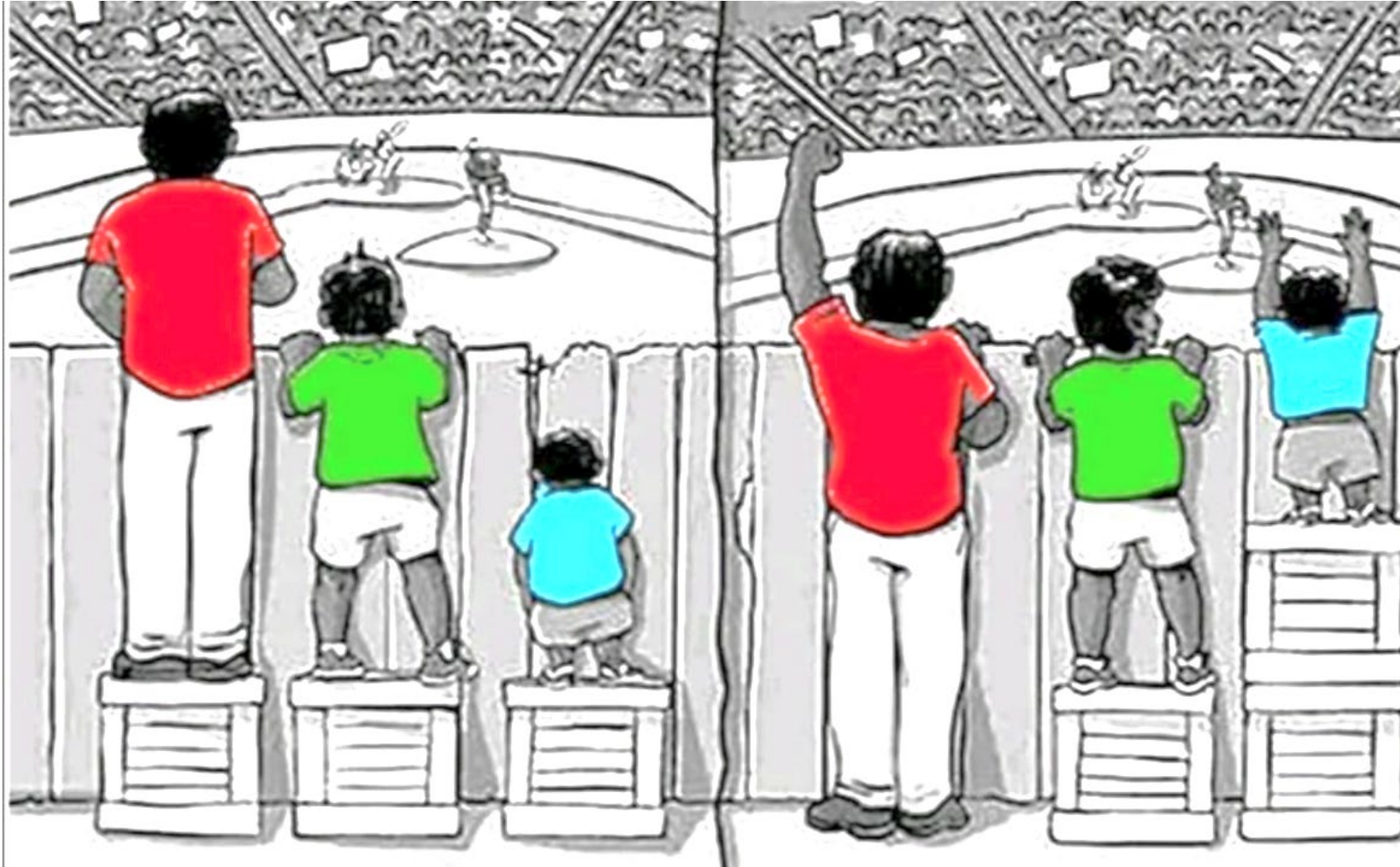
- simulates user reactions and feedback during brainstorming
- generate design ideas, anticipate usability issues, or suggest improvements.

LIMITS

homogeneous outputs, always requiring human supervision.

Pehar, F. AI as Synthetic Users in Human-Centred Design: A Systematic Review.
2025 MIPRO 48th ICT and Electronics Convention, Opatija, Croatia, 2025,
pp. 1500-1509. doi: 10.1109/MIPRO65660.2025.11131969.

One size does not fit for all



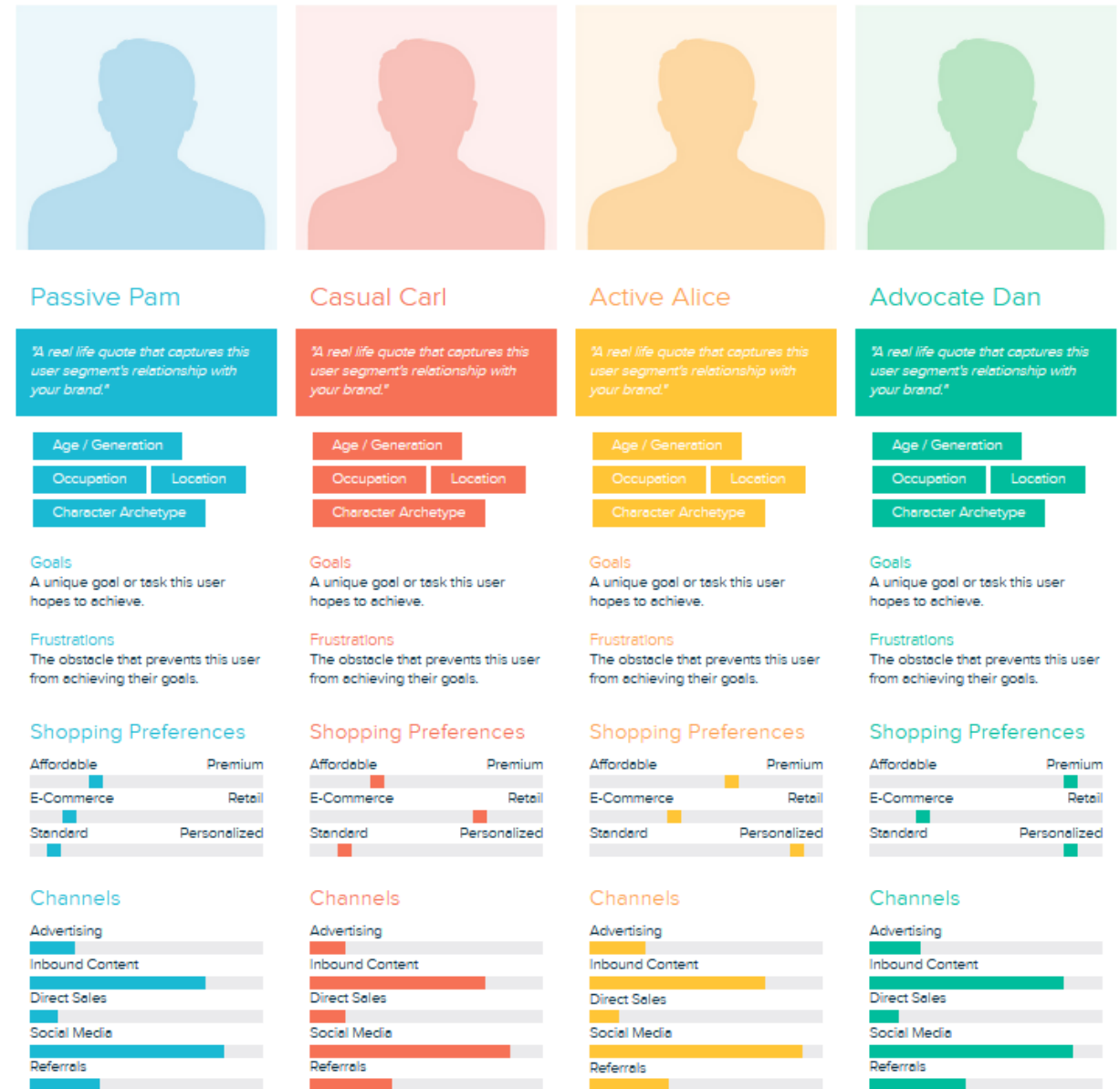
Embrace and implement diversity.

Create more personas to know in depth the different users and beneficiaries, varying on individual and contextual variables.

User personas comparison

. Collection

1. Use Personas with your team members and clients to **deepen your user segments**.
2. Visually compare the user profiles created.



User personas comparison

. Table

1. Use Personas with your team members or clients to **deepen your user segments**.
2. Condense your research and analysis to form an **executive summary**, highlighting the unique characteristics between the segments.

User Persona Comparison



Created by:
Name Last Name
DD.MM.YYYY

Attn:
Name Last Name
Name Last Name

Goals

Explain what you are looking to achieve with this User Persona Comparison. Consider who among your team members or clients will benefit from identifying your user segments.

- ✓ Product Roadmap Decisions
- ✓ Sales Team Goal Setting
- ✓ Content Marketing
- ✓ Social Media Targeting
- ✓ SEO Optimization
- ✓ Pricing Tiers Determination
- ✓ HR and Recruiting
- ✓ Investor Presentations
- ✓ Stakeholder Reporting

Sources

Customer Feedback & Testing

Customer Data

Product Analytics

Social Analytics

Surveys

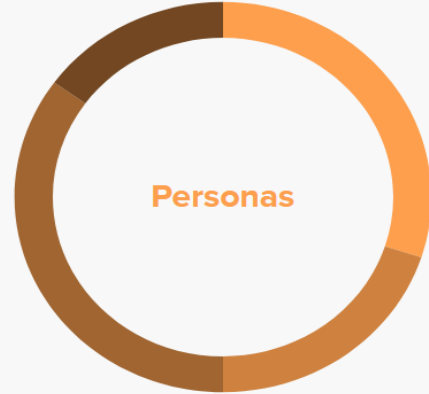
Market Research

Summary of Key Findings

Use this space to condense your research and analysis to form an executive summary.

Why is your audience separated into the following segments? What are some high-level differentiators between each segment? Your segmentation parameters may be based on: demographics, psychographics, geography, usage behavior, levels of engagement, or any other criteria specific to your product, service, or industry. How is each segment distinct and why is each one necessary to target?

Include any supplementary information you feel will help your readers understand your customer segments. When filling out the tables below, concentrate on the unique characteristics between the segments.



Personas

- 30% Passive Pam
- 35% Active Alice
- 20% Casual Carl
- 15% Advocate Dan

Action Items

Based on the findings, list important action items and urgencies related to targeting and communicating with your audience.

0 out of 4 items completed

- ☐ Prioritize content creation for Segment B
- ☐ Push direct sales efforts for Segment C.
- ☐ De-prioritize Segment A's feedback on product iterations.
- ☐ What else?

Create your personas

1. Starting from the Stakeholders analysis, identify the **target**
2. Fill in the Personas portraits (one per profile).

!! 1 personas is not enough:
it is a conceptual and procedural
error.



This will bring you to the
Value proposition of the
project→

PERSONAS	
Demographics (e.g., name, age, role, income, location)	Taks, habits (e.g., digital tools used)
Pain Points & Frustrations (e.g., what they're struggling with)	Goals (e.g., What they want to reach and metrics to assess it)
Data they can produce/use	Expected outputs

Summarise your Personas

Users' Personas Overview		
		
Numbers How many people does the persona Represent	Numbers How many people does the persona Represent	Numbers How many people does the persona Represent
Role and tasks	Role and tasks	Role and tasks
Needs & Goals What to they need and want to reach	Needs & Goals What to they need and want to reach	Needs & Goals What to they need and want to reach
Data they can produce/use	Data they can produce/use	Data they can produce/use
Expected outputs	Expected outputs	Expected outputs

How might we

How might we...?

How Might We (HMW for short) questions are used to inspire ideation and brainstorming solutions to a complex problem. Basically, they are **design challenges framed as questions** that we answer by generating ideas.

Problem Citizens find it dangerous to move around certain areas, in the evening...

Challenge (HMW) How might we increase the security of citizens?

Insight Citizens are unaware of the available resources and are disoriented about the actions to be taken.

Challenge (HMW) How might we guide citizens to «choose» reliable sources (or entries)?

Value proposition

The so-called value proposition is a concise description of **the unique value that the project can offer to meet the interests of stakeholders.**

It explains how the proposed solution responds to specific needs and why it differs from existing alternatives.

EXAMPLE

Sservice available via app,
designed to support **university students and young workers (18-26)** away from home in the search for a housing solution. In particular, the app offers a selection of housing solutions (e.g. apartments, studios, shared flats), described by peers, students and workers, to help those who do not know the city context. This way, people will be able to get to know the city and find the accommodation that best suits their specific needs.

ONE TEMPLATE

The [***service***] addresses [***target***]
that [has/wants/live, miss ... a **specific problem/need**]
by [***main distinguishing feature or capability of the service***] and [other additional peculiar function],
to provide [the unique **value**] and [***benefit for the target***].



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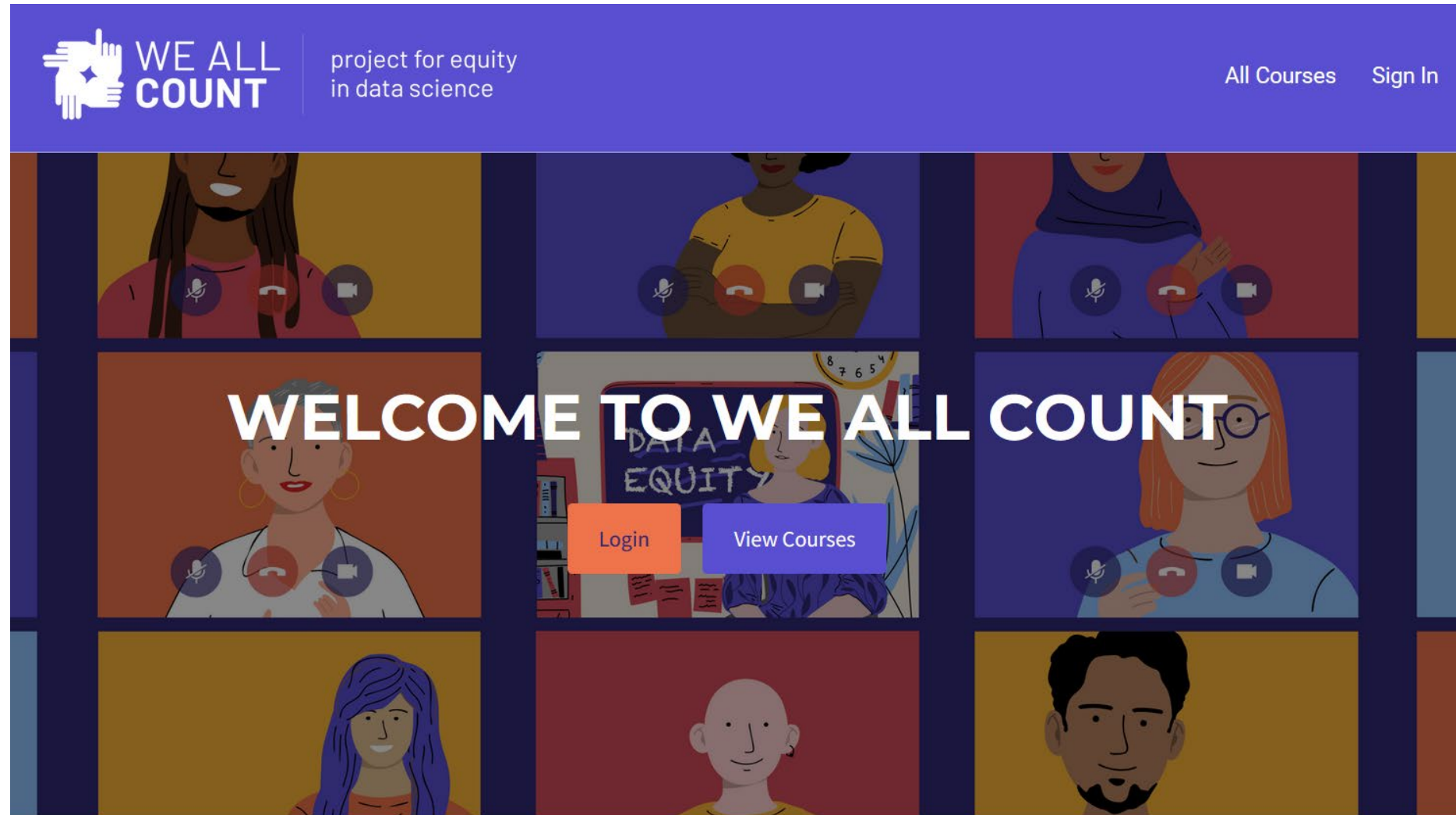
Additional resources

Test test test

AUTHOR		CONTACT DETAILS		FINAL DATE FOR COMMENTS
PRODUCT UNDER TEST What's being tested? What are the business and experience goals of the product?	TEST OBJECTIVES What are the goals of the usability test? What specific questions will be answered? What hypotheses will be tested?	PARTICIPANTS How many participants will be recruited? What are their key characteristics?	TEST TASKS What are the test tasks?	RESPONSIBILITIES Who is involved in the test and what are their responsibilities?
BUSINESS CASE Why are we doing this test? What are the benefits? What are the risks of not testing?		EQUIPMENT What equipment is required? How will you record the data?		LOCATION & DATES Where and when will the test take place? When and how will the results be shared?
PROCEDURE What are the main steps in the test procedure? 				

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A Guide to Data Innovation for Development - From idea to proof-of-concept

NOVEMBER 16, 2017

'A Guide to Data Innovation for Development - From idea to proof-of-concept,' provides step-by-step guidance for development practitioners to leverage new sources of data. It is a result of a collaboration of UNDP and UN Global Pulse with support from UN Volunteers, led by UNDP innovation teams in Europe and Central Asia and Arab States.

<https://www.undp.org/publications/guide-data-innovation-development-idea-proof-concept>